

Exercise 1

Submission date: 25.11.25 (23:59)

Prerequisites

make sure argos3 is installed:

```
argos3 -q all
```

the expected output is a printout of all available plugins (there are a lot).

- if argos isn't installed - refer to https://github.com/OmriPer/pipuck_sim/blob/main/docs/installation.md

Exercise Instructions

Write a controller that drives forwards until there is an obstacle (wall) 10 cm away, then stop. The LEDs should be lit in green while driving, and red while stopping.

Work Space Structure

The `ex1_ws` Repository - https://github.com/OmriPer/ex1_ws contains 3 directories:

- `controllers/` : Contains the source code for robot controllers.
- `experiments/` : Contains the `.argos` XML configuration files for simulations.
- `loop_functions/` : Contains custom loop functions to manage experiment logic.

You only need to write your code under the `controllers/` directory.

Don't modify any other directory - so your setup is identical to the test setup.

Rules

1. Make sure your code is tidy and well-documented.
2. Make sure your ID is written at the top of any file you modify.
3. Name any source you used in your README file.

Files to submit

1. You should submit your project in the same structure as `ex1_ws`.
2. Add a README file with your ID, and a list of any external source you used.
3. **Do not submit any executable files, or any files that can be regenerated.**

Self Test

For self-examination you can:

1. compile your workspace

```
./compile_ws
```
2. run the experiment

```
argos3 -c experiments/config1.argos
```

- if this process fails, it will probably fail in the test environment as well.